

Examiner reports

Q1.

This lengthy question tested a range of skills and gave students the opportunity to show how well they had mastered these. There were few who could not pick up some marks here.

For part (a), most started off with a correct combination of the $x\sqrt{x}$ into a single power and differentiated the expression correctly. It was expected that at this stage they would explain that for a stationary value the gradient was 0. Many simply used this fact.

Here one approach was to solve the equation and select $x = 1$. This approach did the groundwork for part (b). Alternatively, they could verify that $x = 1$ resulted in a gradient of 0. This approach was easier here but meant that more work was needed in part (b).

Use of the second differential was needed to prove that this was indeed a maximum. Finally, it was necessary to show that $x = 1$ gave $y = 3$. Many omitted this last step and so also lost the R1 mark, which required clear mathematical rigour throughout. However, simply failing to explain that for a stationary value the gradient was 0 did not result in the loss of the R mark.

For part (b), as an alternative to algebraic methods, it was permitted for students to use graphical calculators to identify this second stationary point.

Q2.

Almost all students selected the correct response for this multiple choice question. The

most common incorrect answer was $\frac{dy}{dx} = -\frac{2}{x}$, but this was not significant in number.

Q3.

This question saw most students making some progress, with a full range of marks being awarded. A high proportion of students realised that implicit differentiation should be used, but very few differentiated efficiently. Instead, the approach seen most frequently was to first expand the brackets and then differentiate. This resulted in more work to find the required solution. The most frequent mark seen for this question was 7/7. Students who made mistakes at the beginning were still rewarded for using correct techniques.

Q4.

Again, the vast majority of students answered this multiple choice question correctly, giving the answer 7.

Q20.

In general, this question was a good source of marks for many students. Most students

were able to find the correct expression for $\frac{dy}{dx}$ in part (a). In part (b), the majority of students were able to obtain the printed answer in a convincing manner by firstly clearly

indicating that they were finding the value of $\frac{dy}{dx}$ at $x = 2$. There were, however, some cases of students who obtained the printed answer by clearly incorrect methods and some by showing incomplete details; both of which were penalised heavily. In part (c)(i), those

students who correctly approached the problem by trying to solve the equation $\frac{dy}{dx} = 0$ generally picked up 2 of the 3 marks for finding the correct coordinates of M but quite frequently then forgot to state the conclusion, that the stationary point lay on the x -axis.

In a number of solutions, students, having solved $\frac{dy}{dx} = 0$ correctly to get $x = -2$, just stated that the point M lay on the x -axis. In part (c)(ii), those students who produced a rough sketch were usually more successful in recognising that the x -axis was the required tangent than those who used the general equation of a line. In this context, it was not always the case that students realised that the product of zero and $(x + 2)$ was 0. Those students who attempted part (d) generally understood the method of solution but the required coordinates of T were frequently incorrect.

Q21.

In part (a), apart from a small number of students who treated the expression as a product of terms instead of a sum and hence gained no marks, this was usually answered correctly. Where there were errors, it was usually with the first term as the students wrote e^{3x} or $3e^x$ instead of $3e^{3x}$.

In part (b)(i), most students applied the quotient rule correctly. However, poor notation such as $1 + \cos^2 x$ instead of $(1 + \cos x)^2$ or $\sin X^2$ instead of $\sin^2 x$ and sign errors caused marks to be lost. There were also examples of false cancellation leading to the correct result. In part (b)(ii), here the students tended to do either very well or very badly, scoring 2 or 0. Those students attempting to use $y = \ln(\sin x) - \ln(1 + \cos x)$ rarely made progress.

Q22.

- (a) Those candidates who began by finding $\frac{dy}{dx}$ usually found the correct gradient and most gave the equation of the tangent in the required form. A few made arithmetic errors such as $y - 6 = 9x + 9$ leading to $y = 9x + 3$. A few, having found the gradient to be 9, then incorrectly used the negative reciprocal, i.e. $-\frac{1}{9}$. Some candidates did not realise that differentiation was required and attempted to find the gradient using the coordinates of the two points $(-1, 6)$ and $(2, 33)$.
- (b) (i) Provided no arithmetic errors were made, particularly evaluating $2^5 = 64$, then the correct value of k was found from the equation of the curve. A few candidates used the tangent equation and although they found that $k = 33$, no credit was given.
- (ii) Those who used the equation of the line did not always earn the mark as a concluding statement such as “therefore lies on the tangent” was not made. Some worked backwards and found the gradient using $(2, 33)$. A few

candidates confused the requests in parts (i) and (ii) and consequently earned no marks.

- (c) (i) The majority of candidates integrated correctly. Most substituted the limits in the correct order but many sign and arithmetic errors were made. A common error was evaluating $\frac{5}{6} - \frac{54}{6}$ as $\frac{49}{6}$ instead of $-\frac{49}{6}$ and the correct value of the integral was rarely seen.
- (ii) Very few candidates found the area of a trapezium. Most found the area of a triangle instead. Those who used integration in order to find the area under the straight line were far more successful than those who tried to remember the basic formula from GCSE.

Q23.

A majority of students could cope with the powers in part (a) but a significant proportion of others ended up with the wrong second term by thinking that the denominator x should be placed under both the x^2 and the $x^{1/2}$ terms resulting in a common error of the second term being given as $x^{1/2}$ rather than $x^{3/2}$. Most students showed that they could differentiate accurately and a large majority showed that they knew the method for finding the equation of the normal to a curve, with many scoring at least 3 of the 4 marks in part (b)(ii). The final part of the question, which introduced the stationary point, usually gave average students the opportunity to pick up the first mark but in general only the most able students could then cope with the fractional indices to correctly reach $x^{5/2} = 8$ and then to solve this to show that the x -coordinate of the stationary point could be written as $2^{6/5}$.

Q24.

This question was well answered by the great majority of the students.

- (a) Any errors that occurred were usually associated with the second term where $\tan 2x$ was differentiated to $\sec^2 2x$ or $2\sec^2 x$ instead of $2\sec^2 2x$. Attempts at using the quotient rule on $y = \frac{x^4 \sin 2x}{\cos 2x}$ was often prone to errors.
- (b) Most students used the quotient rule correctly, although some had the two terms in the numerator reversed. Those who chose to use the product rule were mostly correct, with a few making a sign error. A small number of students failed to go on substituting $x = 3$ into their derived equation. A common incorrect answer was $\frac{3}{2}$; arising from students expanding the denominator incorrectly to $x^2 - 2x - 1$.

Q25.

In part (a), differentiation had been well drilled and most candidates earned all 4 marks for the two derivatives. Occasionally, the $+1$ was omitted or an extra 5 or $+x$ was included, and a few felt the need to add $+c$ to both their answers.

In part (b), those candidates who substituted $x = -1$ into their expression for $\frac{dy}{dx}$ often obtained a gradient of 12. However, not all used this value: some took the negative

reciprocal and then formed the equation of the normal. Some candidates did not use their expression for $\frac{dy}{dx}$ in order to find the gradient but erroneously used the gradient of AB , with value 2, instead.

Surprisingly, a very small minority earned full marks in part (c). Most candidates verified the nature of the stationary point by considering the sign of the second derivative. However only a handful recognised the need to verify that the point was stationary in the first place, by showing that the first derivative was equal to zero.

In part (d)(i), most candidates were able to integrate the expression, with only the weakest candidates not making any attempt. Poor notation was used, with many including the integral sign after integrating. However dealing with the substitution of -1 caused many errors. Manipulation of fractions caused problems too; it was disappointing to see many candidates with correct fractions being unable to combine these to give a value of 8 or equivalent. Many candidates did not find the actual value of the definite integral until part (d)(ii), and on this occasion full credit was given, but this may not be the case in future.

In part (d)(ii), most candidates calculated the area of the relevant triangle to be 2 and subtracted this from their answer to part (d)(i). Errors included writing the area of the

triangle $\frac{4 \times 2}{2}$ as and careless arithmetic such as $8 - 2 = 4$.

Q26.

This final question proved to be less demanding than has been the case in many of the past papers for this unit. Even the unstructured final part of the question produced a significant number of fully correct solutions from the candidates. Parts (a) and (b)(i) were found to be very straightforward with most candidates scoring heavily. It is worth recording that a number of candidates in part (b)(ii) failed to give sufficient justification for taking the gradient m to be -8 . With the answer given, it was not acceptable to just state that the

gradient is -8 without some clear indication that this value came from the value of $\frac{dy}{dx}$ when $x = 8$. In part (c), the most common loss of a mark was through failing to simplify $\frac{3}{\left(\frac{8}{3}\right)}$ correctly. The error of integrating $x^{\frac{5}{3}}$ to get a term in the form $\frac{7}{10}x^{\frac{7}{3}}$ was penalised more heavily.

In part (d), those who found the area of the triangle by solving the equations of OP and

AP simultaneously to find the coordinates of P and then using $A = \frac{1}{2}bh$ were generally much more successful than those who attempted to use integration, as their limits were frequently incorrect: 0 to 8 in both cases rather than 0 to 3.2 for OP and then 3.2 to 8 for AP . It was also surprising to see a significant minority of candidates finding the lengths of

OP and AP and the angle OPA and then using $\frac{1}{2}ab \sin C$ to find the area of the triangle; although a valid approach it took much more time and usually led to rounding errors. The method for finding the area under the curve was well understood but errors in part (c) frequently resulted in the area under the curve being greater than the area of the triangle: a clear error which should have been spotted by reference to the diagram.

Q27.

In part (a), almost everyone earned the mark.

Part (b) was answered very well. Most candidates who realised the numerator was related to the derivative of the denominator and hence required a $\ln$ function were successful in obtaining full marks. There were two very common errors:

- candidates losing the final accuracy by writing $\frac{1}{6} \ln 91 - \frac{1}{6} \ln 21 = \frac{1}{6} \ln 70$
- candidates losing 2 marks by obtaining an incorrect value of k ; the most common value was 6 although $\frac{1}{2}$ and $\frac{1}{3}$ were also seen.

The major errors seen were expressions such as $\frac{2x^2 - 1}{12x^2 - 6} \ln(4x^3 - 6x + 1)$. Some candidates were able to recover by simplifying to the correct expression $\frac{1}{6} \ln(4x^3 - 6x + 1)$ before substitution of the limits; however some candidates substituted for 3 and 2 through the whole unsimplified expression and therefore gained no credit.

Q28.

In part (a), it was essential to use the quotient rule, as requested – it is specifically mentioned in the specification – and to show sufficient steps in the proof of the given answer. It was surprising that quite a number of candidates were unable to go from $\frac{-\cos \theta}{\sin^2 \theta}$ to the final expression.

In part (b), it was good to see some completely correct solutions though some used 30° and 45° , which was incorrect. A substantial number of candidates who correctly started the question lost a minus sign in their work.

Many candidates fell at the first hurdle by not replacing dx by an expression in θ and $d\theta$.

There were B marks available for a correct $\frac{dx}{d\theta}$, for $\sqrt{\operatorname{cosec}^2 \theta - 1}$ converted to $\sqrt{\cot^2 \theta}$ and for the limits in radians, but even these marks were often not earned. Unfortunately, many confused $\frac{dx}{d\theta}$ with $\frac{d\theta}{dx}$ and of course made no headway after that. Candidates who produced the correct numerical answer from their calculators without correct working did not gain marks.

Q29.

In part (a)(i) of this calculus question, where students were asked to verify that a given expression was equal to zero when $x = 2$, a minority of students failed to give adequate written evidence, particular when the order of calculation was important; the minimum examiners accepted here was $3 \times 4 - \frac{4}{4} - 11 = 0$, although $12 - 1 - 11 = 0$ was more

convincing. In part (a)(ii) the differentiation of $-\frac{4}{x^2}$ was a hurdle not always overcome by the weaker students. However, those who first re-wrote it as $-4x^{-2}$ usually went on to differentiate it correctly. Many students scored the mark for giving a correct reason in part (a)(iii). In part (b) a large number of the students did realise that they needed to integrate the given expression for the gradient to find the equation of the curve but a significant number of them did not include the arbitrary constant which is essential here, and only a minority knew how to find its value from the given information. Although only a minority, it was still disappointing to see more students than expected writing down the equation of the curve in the form $y - y_1 = m(x - x_1)$.

Q30.

This question was in general a good source of marks for the majority of students. The main cause of loss of marks was a common desire to use an approximate numerical value for e .

Q31.

In part (a) it was good to see almost everyone being able to differentiate correctly.

A few then went on to divide their expression for $\frac{dy}{dx}$ by 2 or 3 or 6, but this was not penalised at this stage.

In part (b) most candidates tackled the verification that P was a stationary point first and generally did so correctly, although some did not present accurate working, writing things such as $18 + 6 - 12 = 0$. Many stopped at this point, although a few tried listing

various values of x with the corresponding value of $\frac{dy}{dx}$, and occasionally discovered

the other value of x for which y was stationary. Many candidates equated $\frac{dy}{dx}$ to zero, but not all were then able to factorise the quadratic correctly. Some felt the need to use the quadratic equation formula and were not always successful in finding the correct solutions.

Part (c)(i) Those who had the correct expression for $\frac{dy}{dx}$ usually found the second

derivative correctly. Many ignored the request to find the value of $\frac{d^2y}{dx^2}$ when $x = -1$, but this was sometimes generously credited in part (c)(ii), where it was pleasing to see most candidates being able to interpret the sign of the second derivative in order to describe the nature of the stationary point.

Q32.

In part (a)(i) candidates who recognised that the first step required was differentiation generally produced correct answers, but a few had trouble with the “ $-x$ ” term and others were unable to combine $-1 + (-4)$ correctly. Some weaker candidates assumed that they could find the gradient using the coordinates of two points.

In part (a)(ii) once again there was much confusion between parallel and perpendicular lines, with many candidates giving the wrong gradient of the tan gent. To earn the final mark it was essential to express the equation in the specified form and some failed to do so; for instance, an answer such as $y + 5x = 17$ is not in the correct form.

In part (b)(i) the integration was generally correct and most candidates made an attempt to substitute the correct limits. A few however could not integrate “14” and left it as a constant. Very few were able to combine their fractions correctly and consequently only a small proportion of the candidates found the correct value of the integral.

In part (b)(ii) candidates who had only found the definite integral in part (b)(i) were generously rewarded with the remainder of the marks here. Not everyone recognised the need to find the area of a triangle in order to find the area of the shaded region. A few correctly found the area under the line $y = 4x + 8$ by integration, but this is not the recommended approach.

Q33.

Most candidates replaced $\frac{8}{x^4}$ by $8x^{-4}$ and differentiated correctly to score full marks for part (a).

Some other candidates failed to score the accuracy mark because they included a ‘+c’ in their

answer. A very large majority of candidates realised that the value of $\frac{dy}{dx}$ at $x = 1$ was required in part (b), but a small minority did not appreciate that the value of y was also required. Although many candidates found the equation of the tan gent correctly, it was not uncommon to see others finding the equation of the normal instead.

In part (c), a majority of candidates equated their expression for $\frac{dy}{dx}$ to zero and attempted to solve the resulting equation, although only about half of the candidates obtained the correct coordinates of M . Those who had wrong powers of x in their answer to part (a) were restricted to just two marks in part (c). The most common incorrect methods seen either involved solving

$\frac{d^2y}{dx^2} > 0$ or looking for a link between the tangent in part (b) and the stationary point.

In part (d)(i), a high percentage of candidates were able to integrate $\frac{8}{x^4}$ correctly, but errors in the integration of $(x + 3)$ or the omission of the constant of integration resulted in a significant proportion of these candidates failing to score the final mark. Since answers to part (d)(ii) required explicit evidence that the answer to part (d)(i) had been used; the few candidates who just wrote down the answer without any other working gained no credit. However, most candidates applied the ‘Hence’, showing the values 2 and 1 substituted into their answer to part (d)(i) and carrying out the relevant subtraction. However, a

significant number of candidates made arithmetical errors in carrying out the subtraction and so did not score the accuracy mark.

Only the more able candidates appreciated what was required in the final part of the question. The common wrong answers included 0 (presumably from the x -axis being a tangent), 12 (the y value from part (b)) and 5.5 (forgetting to consider the direction of the translation).

Q34.

Part (a) was well answered by the majority of candidates with many fully correct responses seen. Most candidates used substitution of $u = (x^3 - 1)$ to give $y = u^6$. Although $(x^3 - 1)^5$ was usually seen, errors did occur with the associated function of x .

Part (b)(i) was very well answered, with a majority of candidates earning both marks. The major error was from candidates who did not treat this as a product and simply differentiated x to get 1

and $\ln x$ to get $\frac{1}{x}$.

Part (b)(ii) Again, this part was very well answered. Many fully correct responses were seen. The main error was with candidates who wrote the gradient as $1 + \ln x$ which earned the method mark. Then this was often evaluated as $1 + \ln e = 1$ or left unsimplified. Several candidates who

obtained $\frac{dy}{dx} = 2$ then went on to use a gradient of $-\frac{1}{2}$.

Q35.

Many candidates did not have enough room to complete this question in the space available, often because of over-elaborating their answer to part (b)(ii).

Part (a) was answered correctly in the majority of cases.

Part (b)(i) was also done well, with only a small number making slips with signs, or square-rooting when they should have squared.

Part (b)(ii) had only 1 mark allocated and a request to "write down" the required equation. Although approximately 25% of the candidates did write down the correct equation, $x = 4$, of the normal at the maximum point, a far more popular but wrong answer was $y = 8$. Many other candidates produced half a page of working to reach an equation of a line which was neither vertical nor horizontal.

It was pleasing to see a much better response in part (c)(i) with many more finding the correct equation of the normal at P , although many failed to reduce it to a form with positive integers a, b

and, in particular, c . It was not uncommon to see the final answer left as ' $2x + 3y = \frac{99}{4}$ ', just one

step away from the correct answer ' $8x + 12y = 99$ '. In general, only the more able candidates, who made good use of the given diagram to show that the normal at M was

vertical, gained credit for their answers to part (c)(ii) of the question.

